

TCP/IP Model, Encapsulation and Network Addressing

Mujahid Tabassum

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Agenda(Part 1)

- What's a protocol suite/stack
- TCP/IP and the benefits of "Layered Model"
- Data Encapsulation
- Accessing resources/data on a network
- Logical addresses and Data Link (physical) addresses

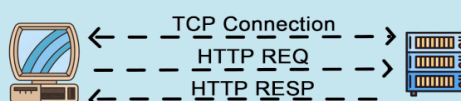
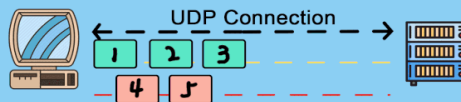
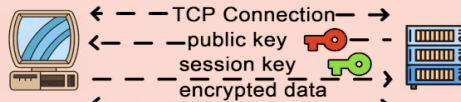
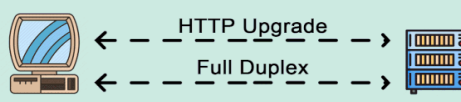
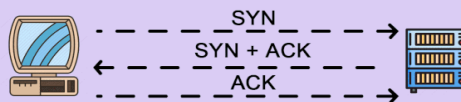
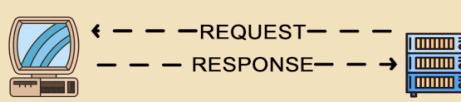
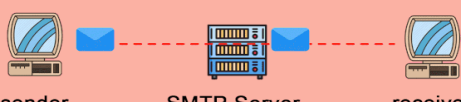
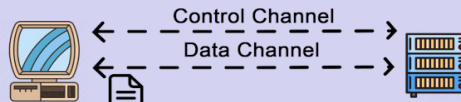


Network Protocol Overview

Network protocols define a common set of rules.

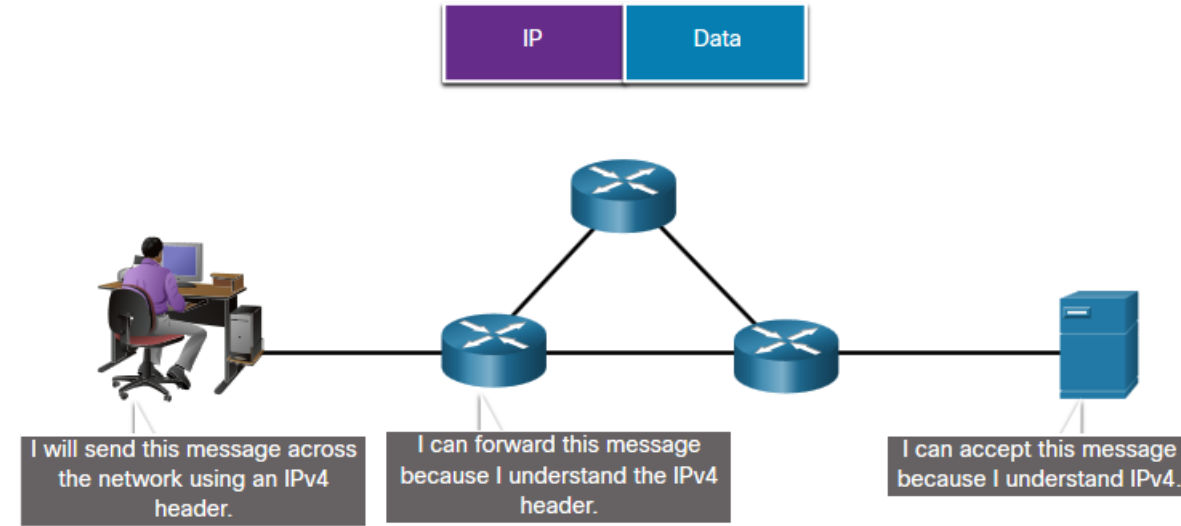
- Can be implemented on devices in:
 - Software
 - Hardware
 - Both
- Protocols have their own:
 - Function
 - Format
 - Rules

Protocol Type
Network Communication Protocols
Network Security Protocols
Routing Protocols
Service Discovery Protocols

8 Popular Network Protocols		
blog.bytebytego.com		
Protocol	How does It Work?	Use Cases
HTTP		Web Browsing
HTTP/3 (QUIC)		IoT Virtual Reality
HTTPS		Web Browsing
WebSocket		Live Chat Real-Time Data Transmission
TCP		Web Browsing Email Protocols
UDP		Video Conferencing
SMTP		Sending/Receiving Emails
FTP		Upload/Download Files

Network Protocol Functions

- Devices use agreed-upon protocols to communicate .
- Protocols may have may have one or functions.
- Network Protocols define:
 - Format
 - Message order
 - Actions on transmission/receipt



Function	Description
Addressing	Identifies sender and receiver
Reliability	Provides guaranteed delivery
Flow Control	Ensures data flows at an efficient rate
Sequencing	Uniquely labels each transmitted segment of data
Error Detection	Determines if data became corrupted during transmission
Application Interface	Process-to-process communications between network applications

Protocol Suites & Standards

- A Protocol Suite is a group of protocols designed to work together
- Typically use open, widely used protocols.
 - Example: Wifi, HTTP, FTP, TCP, IP...
- Protocol Standards established by Institute of Electrical and Electronics Engineers (IEEE) or the Internet Engineering Task Force (IETF)
- Protocol suites based on open standards ensures that products from different manufacturers can work together for efficient communications



IEEE

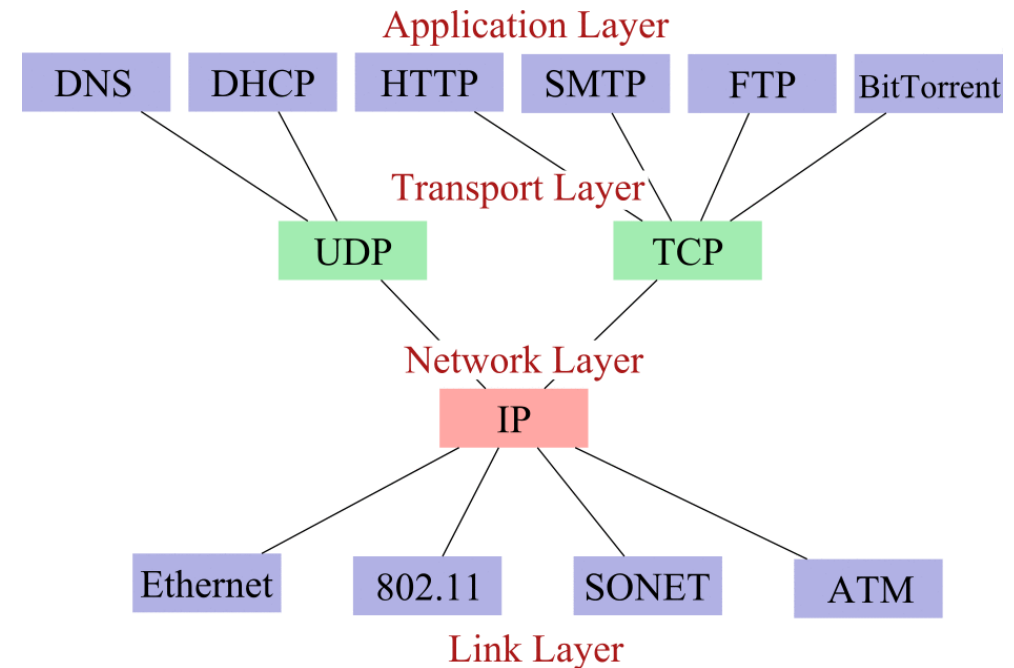
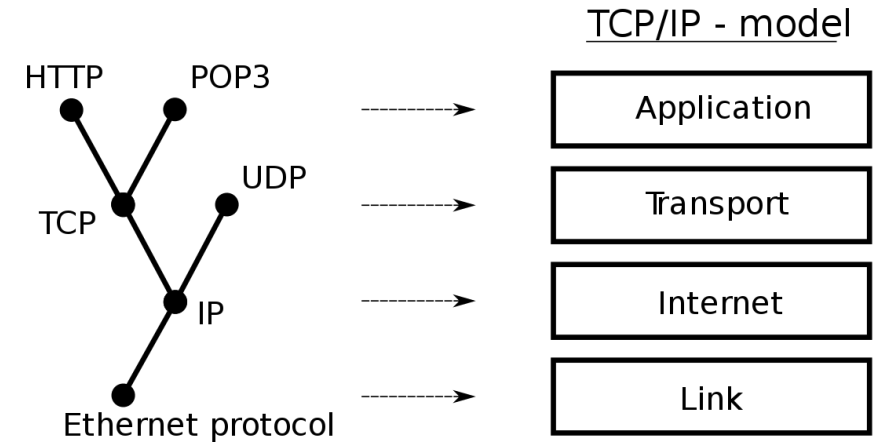
*Advancing Technology
for Humanity*



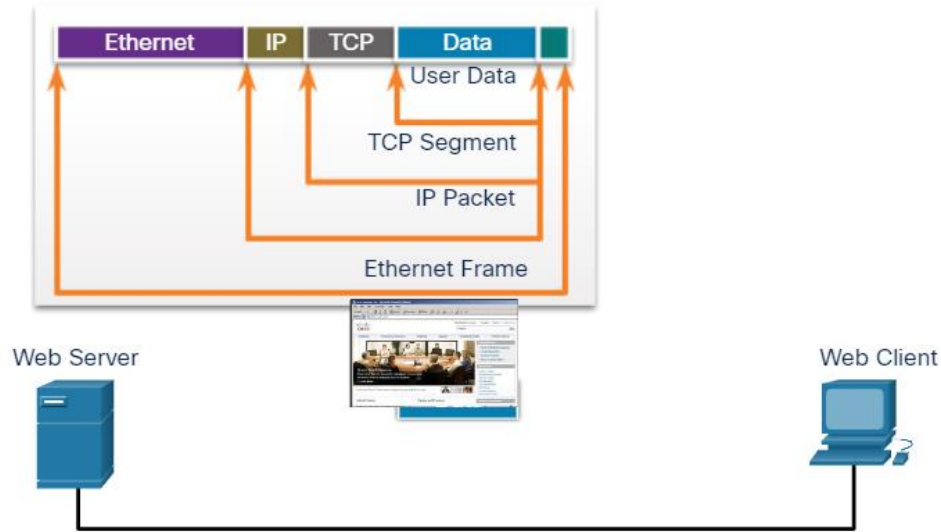
IETF®

TCP/IP Protocol Suite

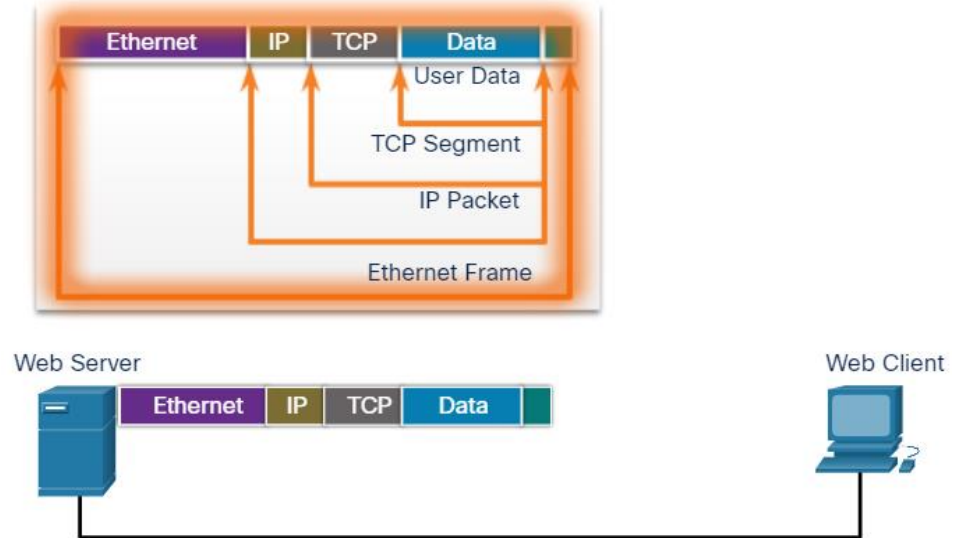
- A whole suite of protocols, including TCP, IP, UDP, ARP, DNS, HTTP, ICMP and many more acronyms!
- Often referred to as TCP/IP, TCPIP, or just IP
- **QI:** The Advanced Research Projects Agency Network (ARPANET) was the first wide-area packet-switched network with distributed control and one of the first networks to implement the TCP/IP protocol suite. Both technologies became the technical foundation of the Internet.
- TCP/IP is now the “standard” protocol suite for the internet



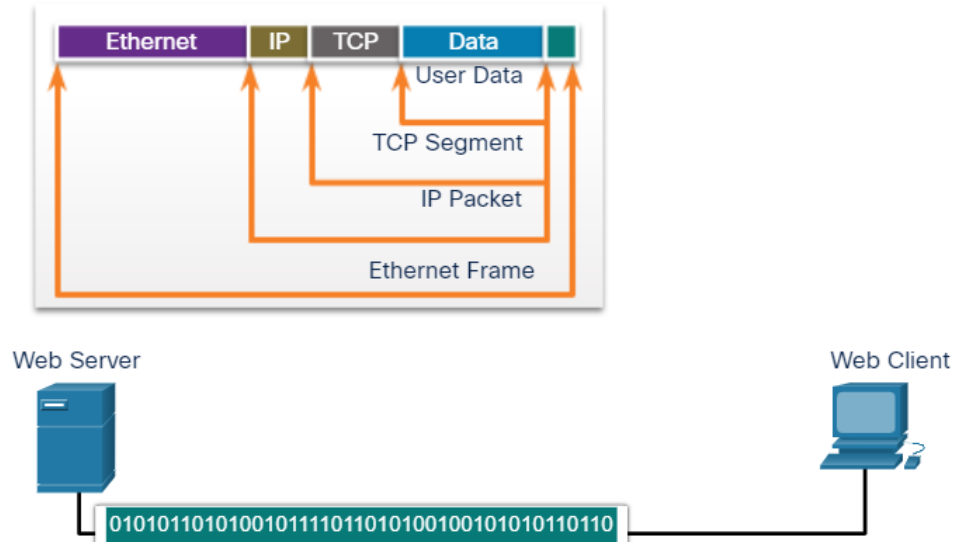
TCP/IP Communication Process



Step 1: Web Server response to user request and sends WEBPAGE data

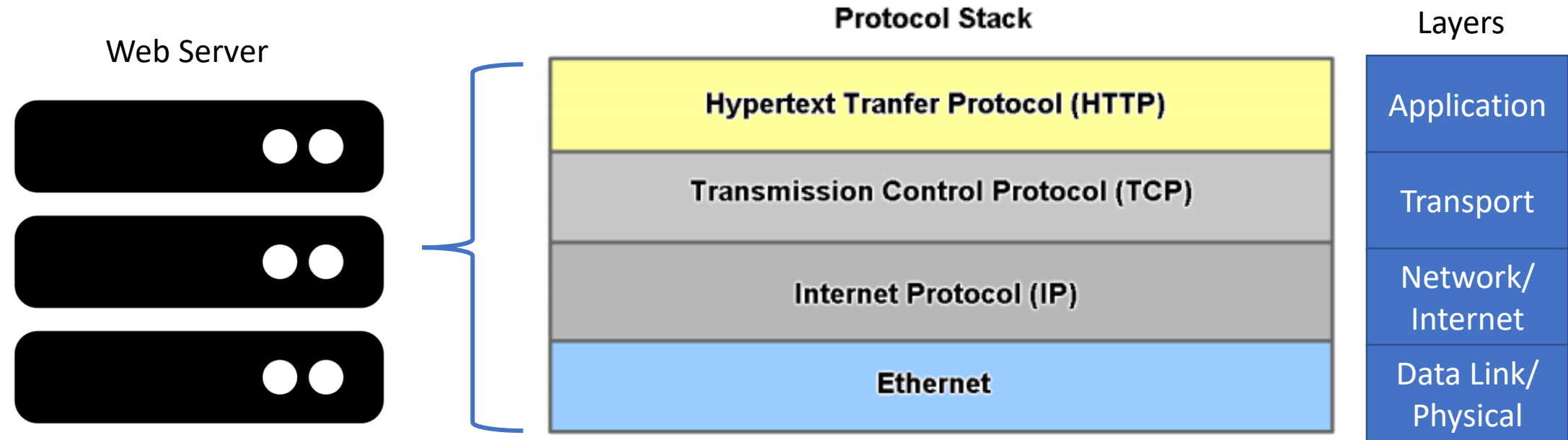


Step 2: Protocols header are added to secure, encapsulate and transmit DATA



Step 3: Data with header is converted to BINARY Number and transmitted over medium

Example: Layered Model Network Comms

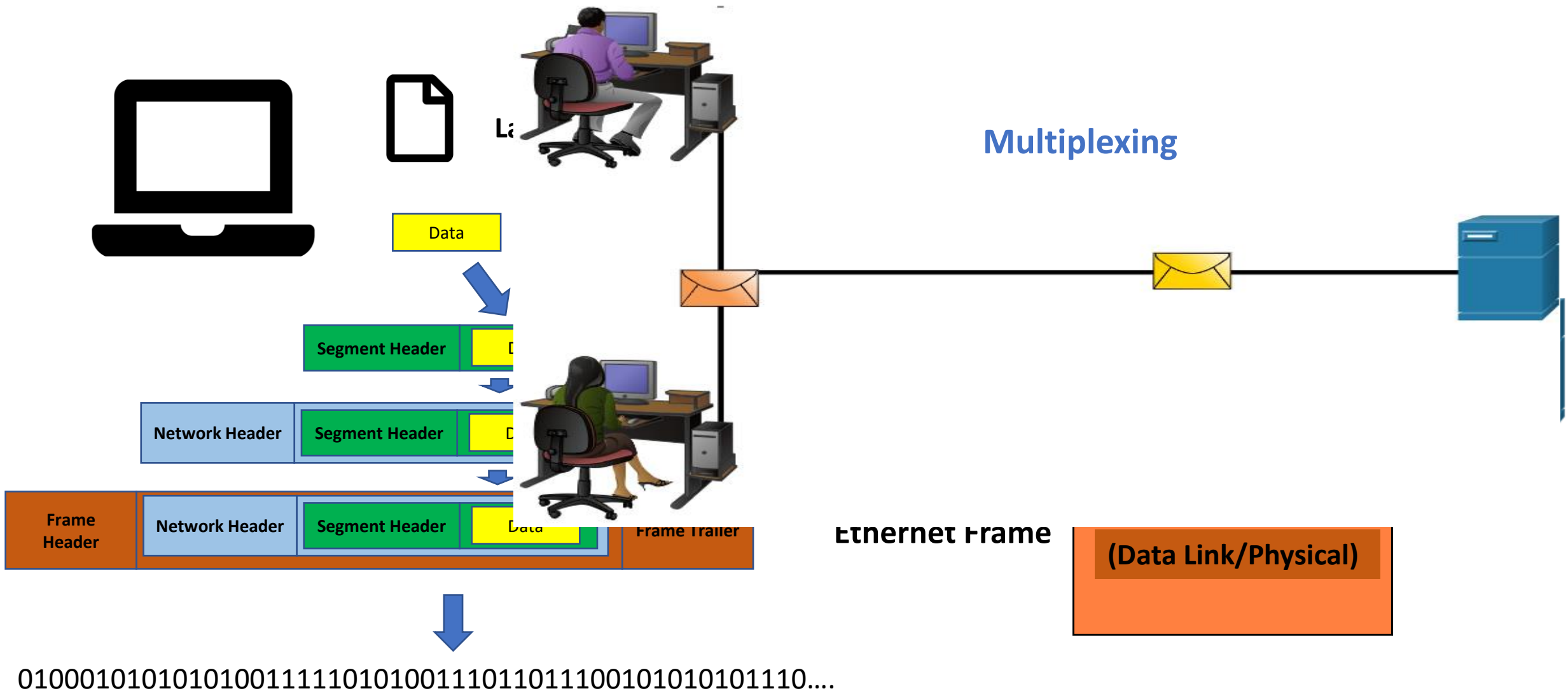


Benefits of Layered Model:

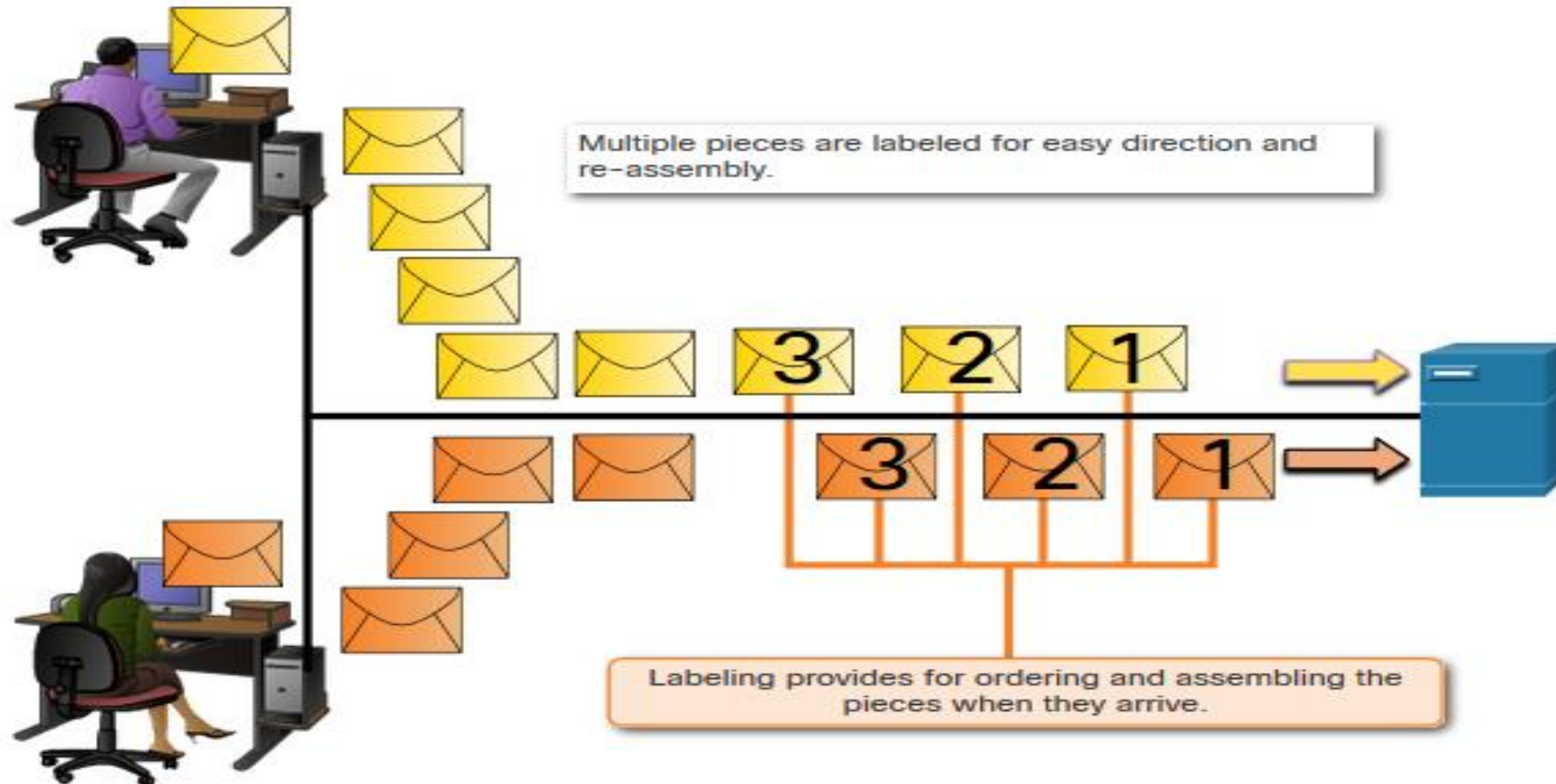
- assists in protocol design
- fosters competition
- changes in one layer do not affect other layers
- provides a common language

The Layered approach breaks down the communication process into smaller and easier to handle interdependent categories, with each solving an important and distinct aspect of the data exchange process.

Comms in TCP/IP: Data Encapsulation



Data Encapsulation - Sequencing

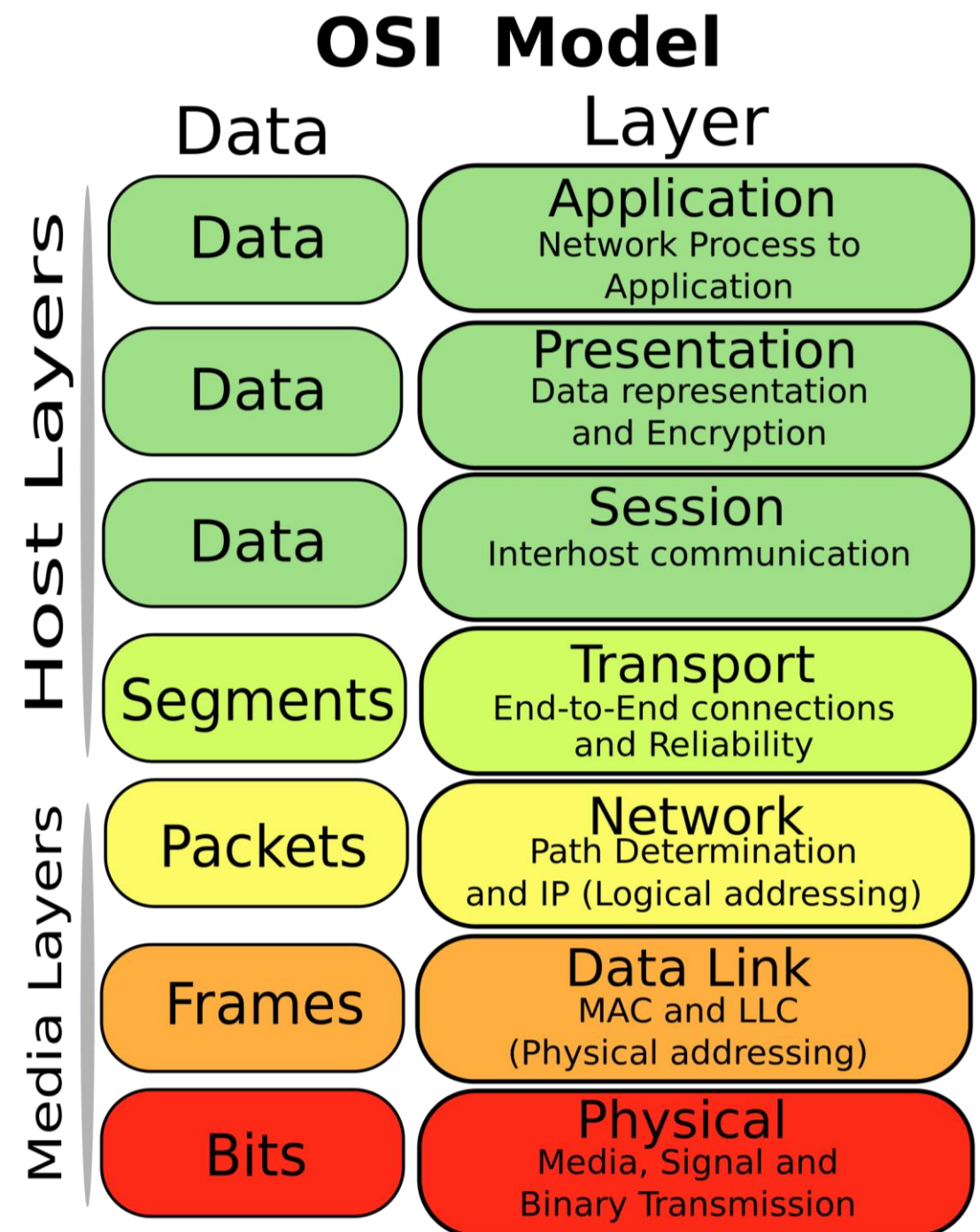


- Sequencing messages is the process of numbering the segments so that the message may be reassembled at the destination.
- TCP is responsible for sequencing the individual segments.

OSI Model

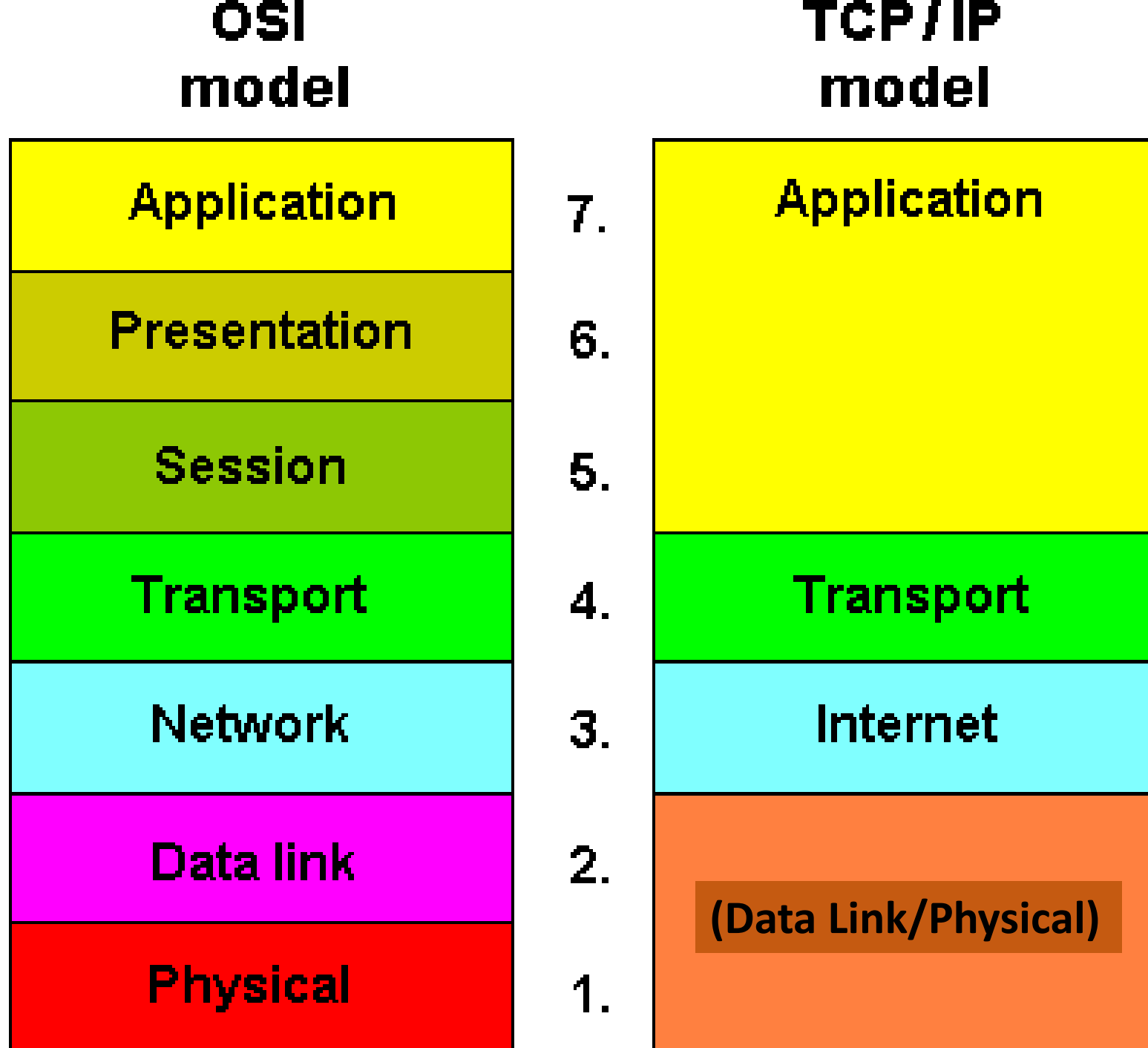
- Open Systems Interconnection(OSI) introduced by the International Organization for Standardization (ISO) in 1984
- Provide a reference model to make sure products of different vendors would interoperate in networks.
- A layer in the OSI model communicates with three other layers:
 - the layer above it, the layer below it, and the same layer at its communication partner.

Image Source: <https://upload.wikimedia.org/wikipedia/commons/thumb/4/47/Osi-model-jb.svg/2000px-Osi-model-jb.svg.png>

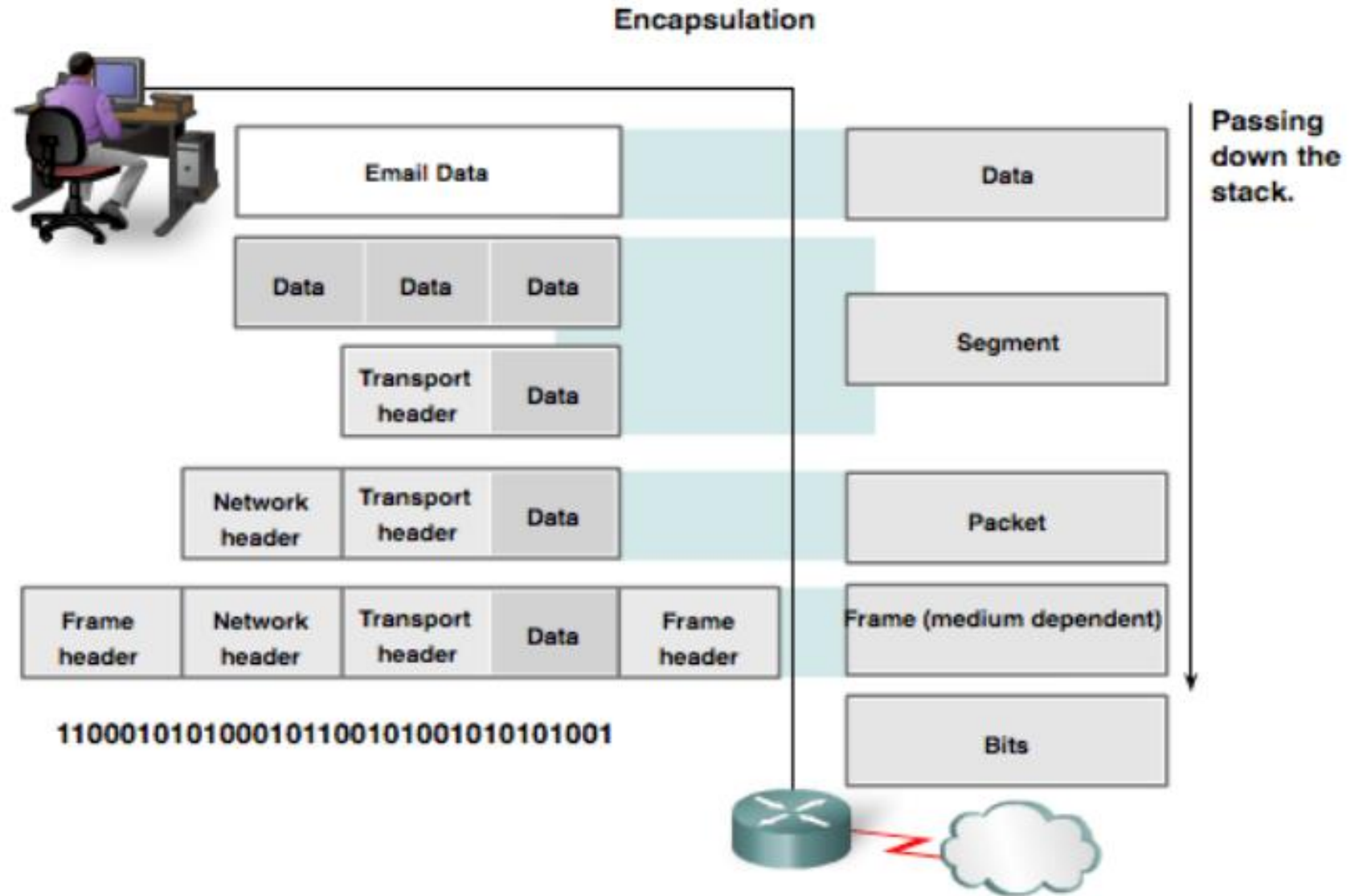


OSI vs TCP/IP

- TCP/IP model combines the presentation and session layer into its application layer.
- TCP/IP combines the OSI data link and physical layers into the host-to-network/network access layer
- Internet developed on around TCP/IP protocols.
 - Thus very popular....
Networks are not usually built on the OSI model, even though the OSI model is used as a guide.



Data
Encapsulation:
Protocol Data
Units (PDUs)

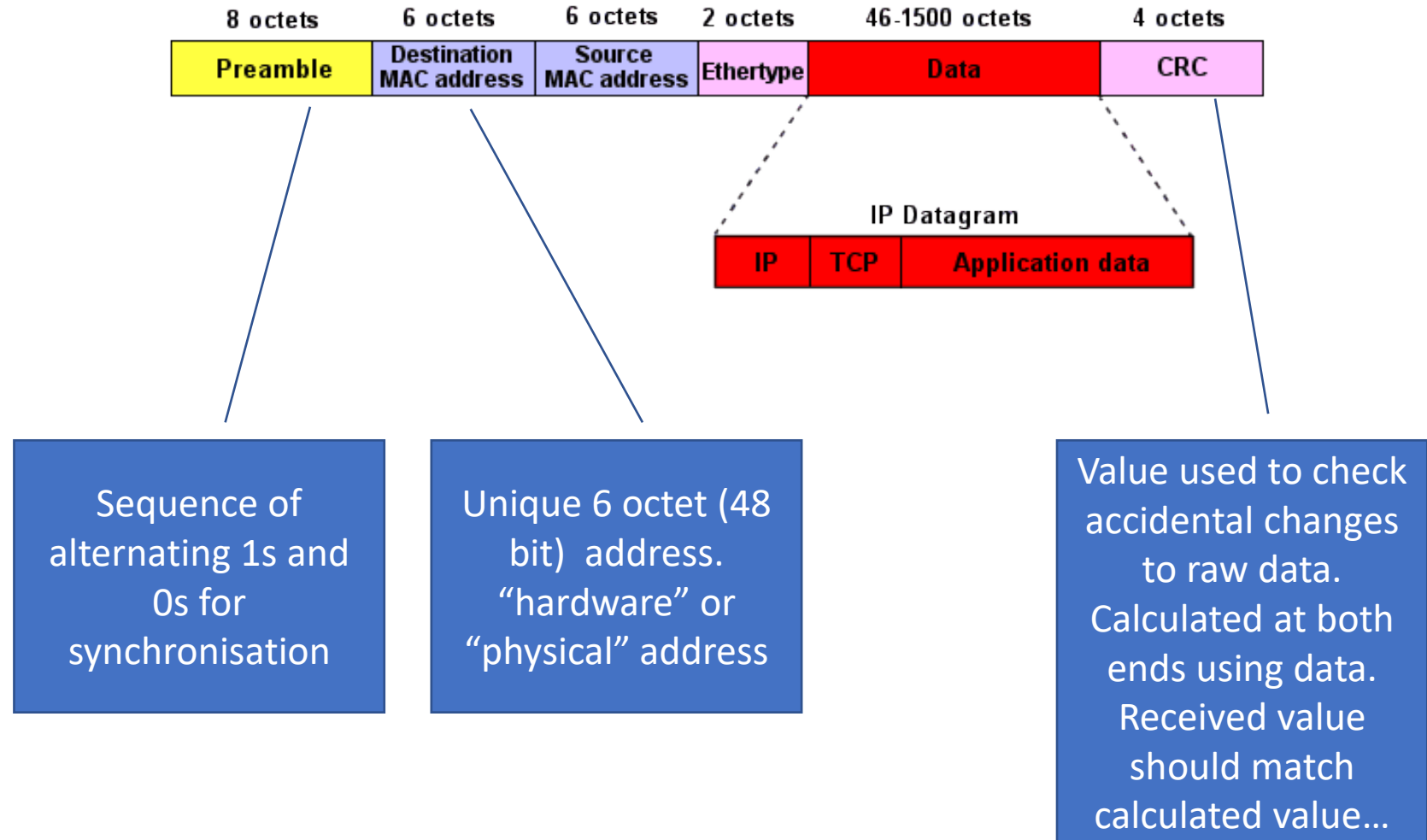


TCP/IP Layers Overview

From the bottom up....

Data Link Layer Communication

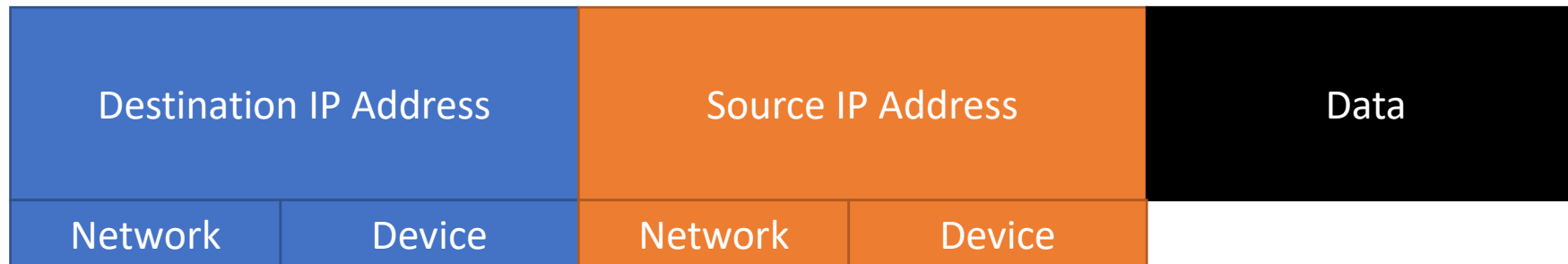
- Data Link Layer protocols transfer data between nodes on a network across the physical layer.
- Supported by many physical layer implementations
 - Wireless/wired
- Octet: represents an 8-bit quantity
- Protocol Data Unit is the **frame**



Internet/Network Layer Communication

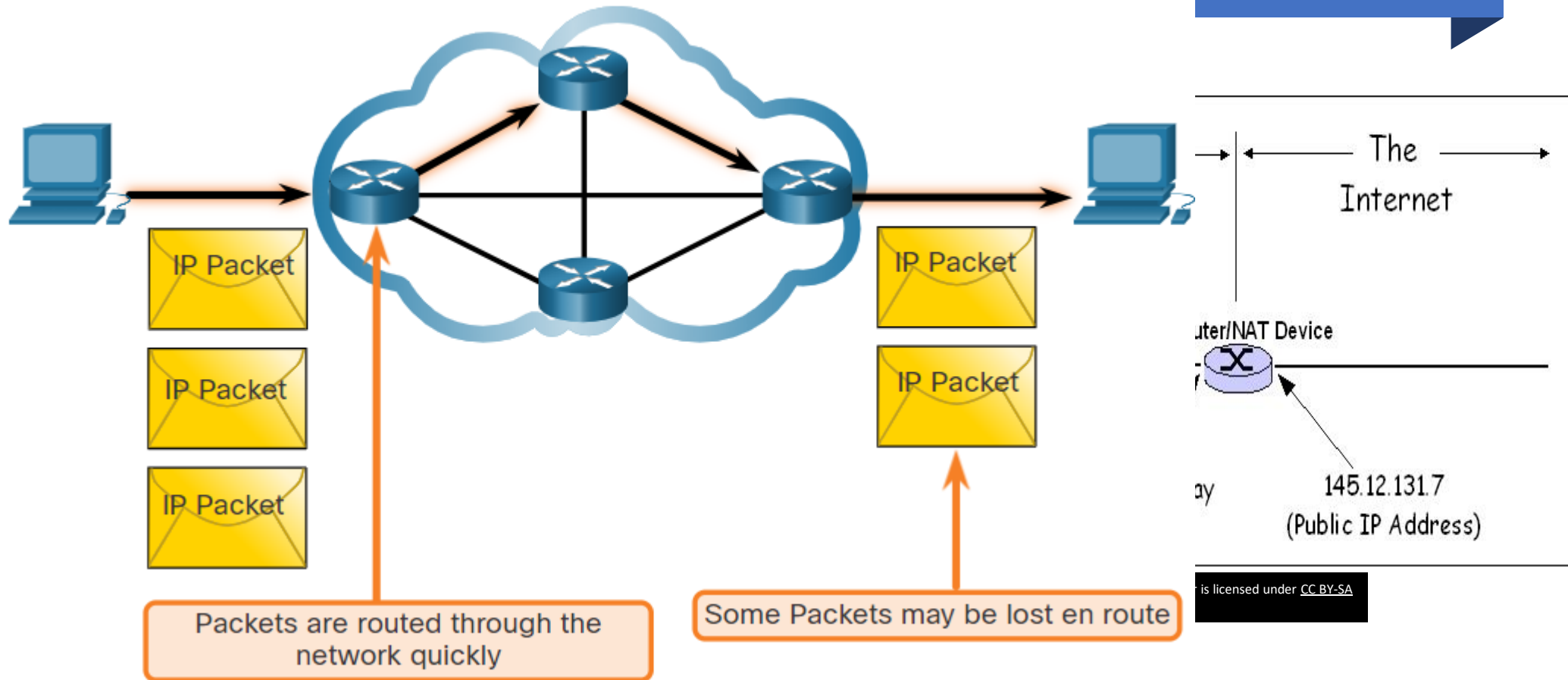
- Network layer protocols primary function is to move data from one network to another network
- Network addresses(IP Addresses) must have a mechanism to locate hosts on different networks
- Intermediary network devices such as routers, decapsulate frames to read destination host address contained in the packet header.
- Routers use the network portion of this address to determine which path to use to reach its destination.
- PDU is the Packet

IP/Network Packet



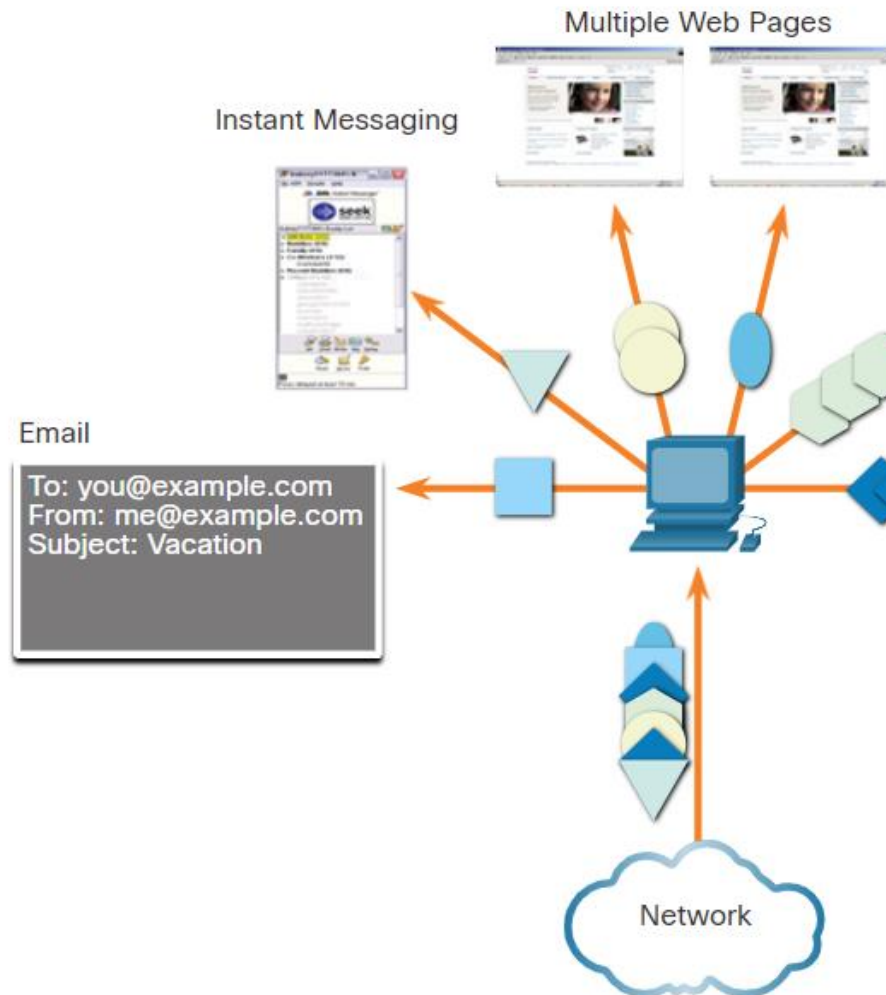
Internet/Network Layer Communication

- IP address
- C
- Includes Host
- Each a unique
- Public assign
- Private inter
- address



Transport Layer Communication

- Transport layer data from the
- How does a c differentiate l
 - Port Num represent
- When a device determine wh destination
- There are gen
 - What's tl
 - HTTP ser
 - FTP servi



Key responsibilities:

- Tracking individual conversations
- Segmentation Data and Reassembling Segments
- Add Header Information
- Identifying the Applications
- Conversation Multiplexing

Application Layer

- Provides access to network.
- Used to develop networked/connected applications.
- Examples include:
 - HTTP – Hypertext Transfer Protocol
 - FTP – File Transfer Protocol
 - SSH – Secure Shell
 - MQTT - MQ Telemetry Transport

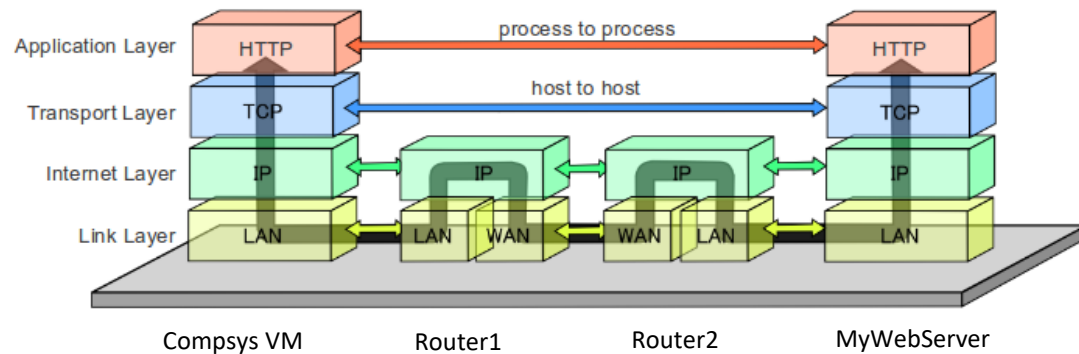


Accessing Resources on a Network

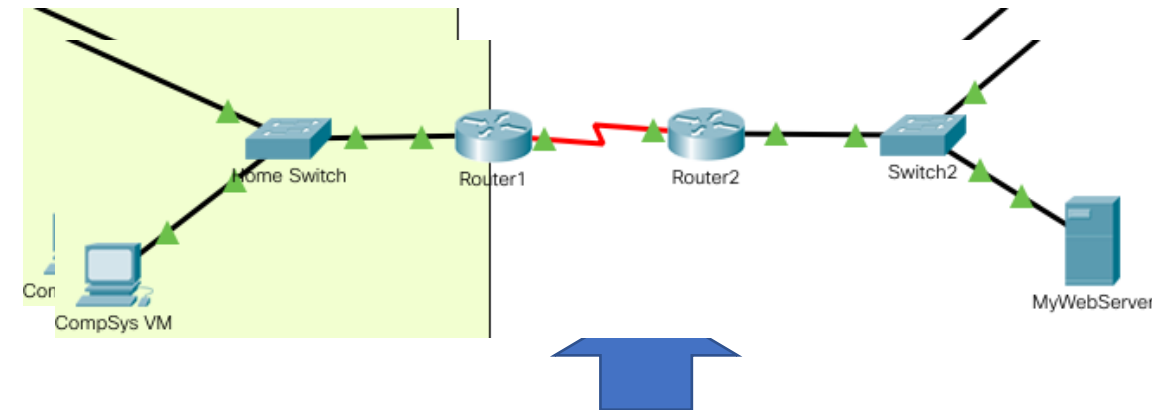
Using TCP/IP protocol stack

Example of Layered Model: Accessing Web Page...

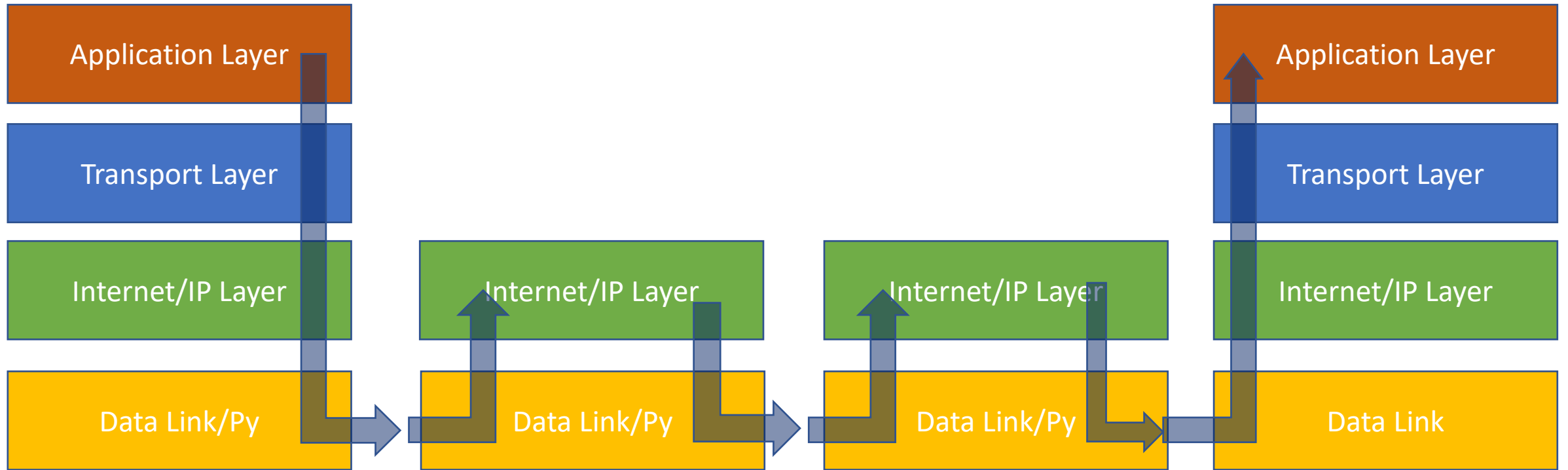
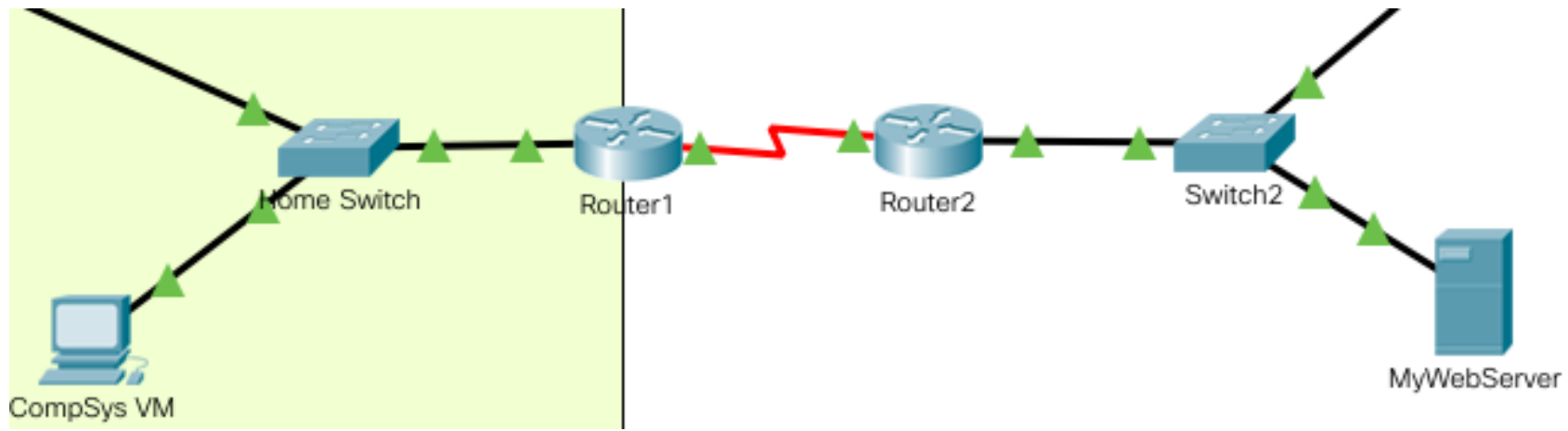
Layered Model



Logical Topology



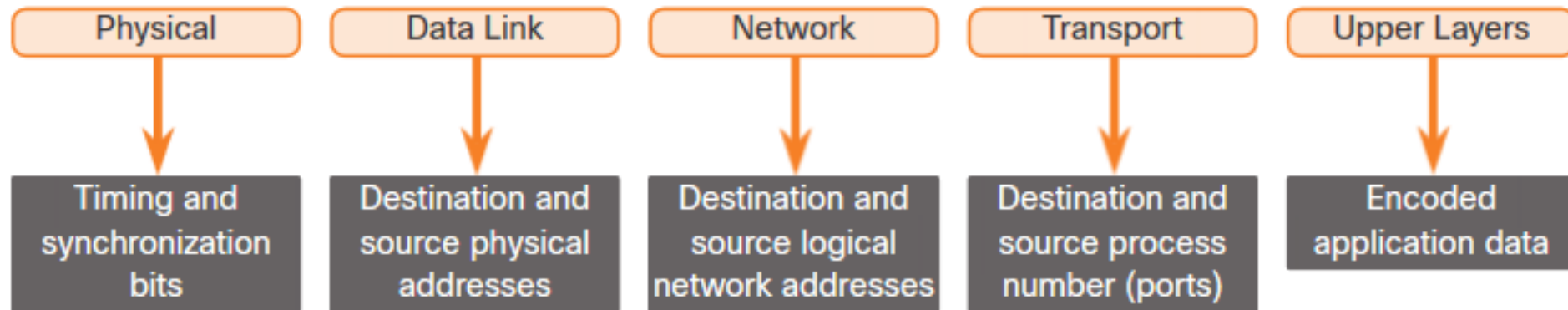
Packet Tracer Example File on Website



Network Addressing

The data link and network layers use addressing to deliver data from source to destination.

- **Network layer source and destination addresses** - Responsible for delivering the IP packet from original source to the final destination.
- **Data link layer source and destination addresses** – Responsible for delivering the data link frame from one network interface card (NIC) to another NIC on the same network.

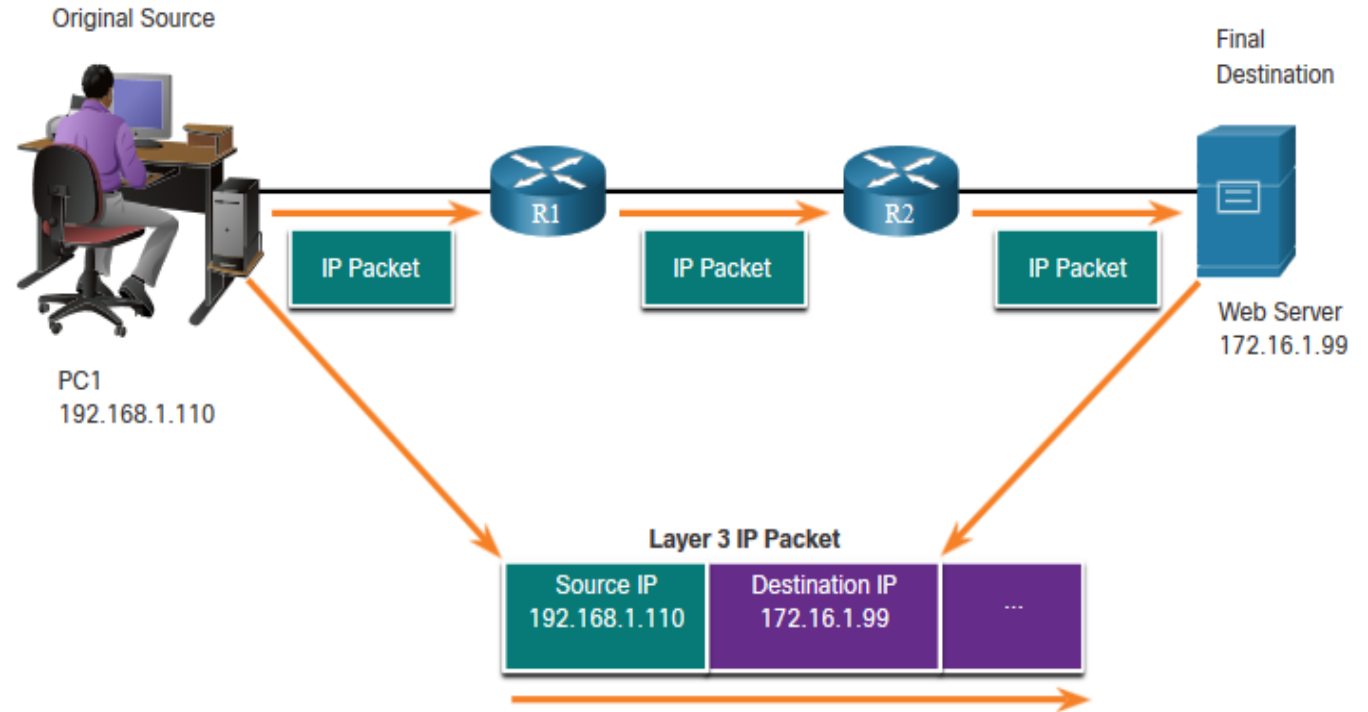


Network Layer: Logical Address

The IP packet contains two IP addresses:

- **Source IP address** - The IP address of the sending device, original source of the packet.
- **Destination IP address** - The IP address of the receiving device, final destination of the packet.

These addresses may be on the same link or remote.



Network Layer Logical Address (Cont.)

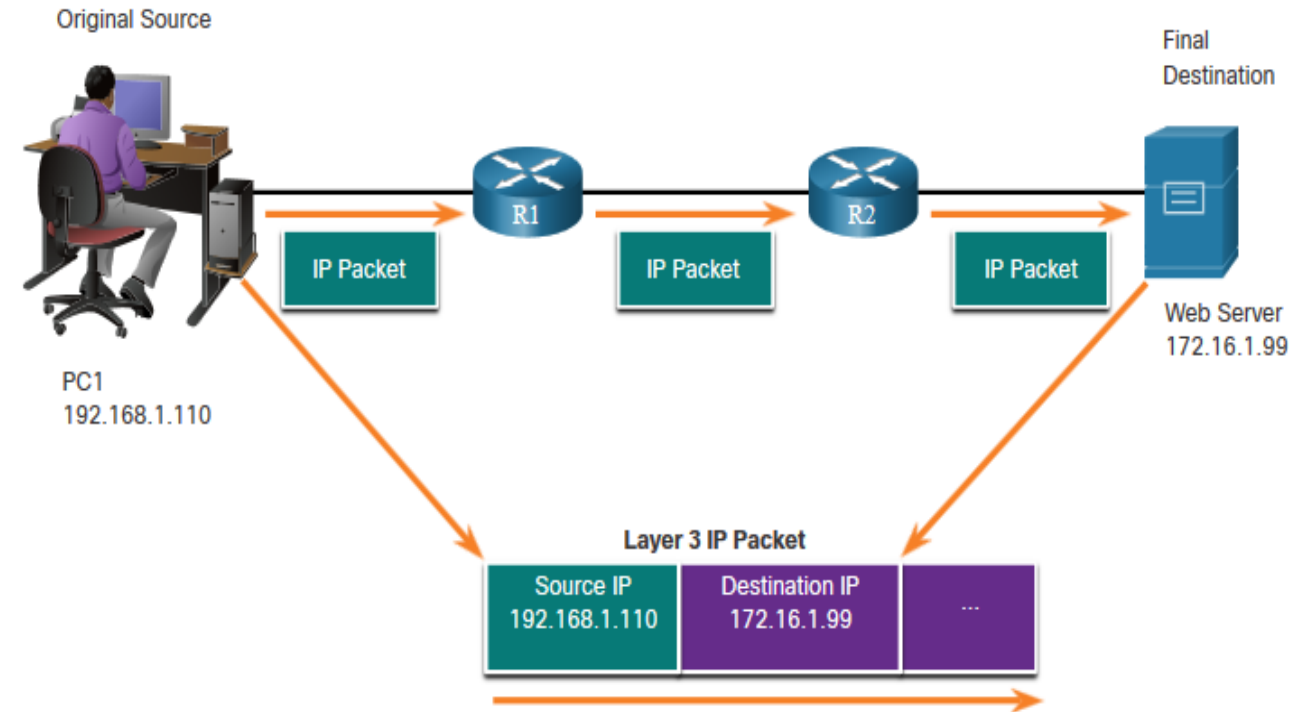
An IP address contains two parts:

- **Network portion**

- The left-most part of the address indicates the network group which the IP address is a member.
- Each network will have the same network portion.

- **Host portion**

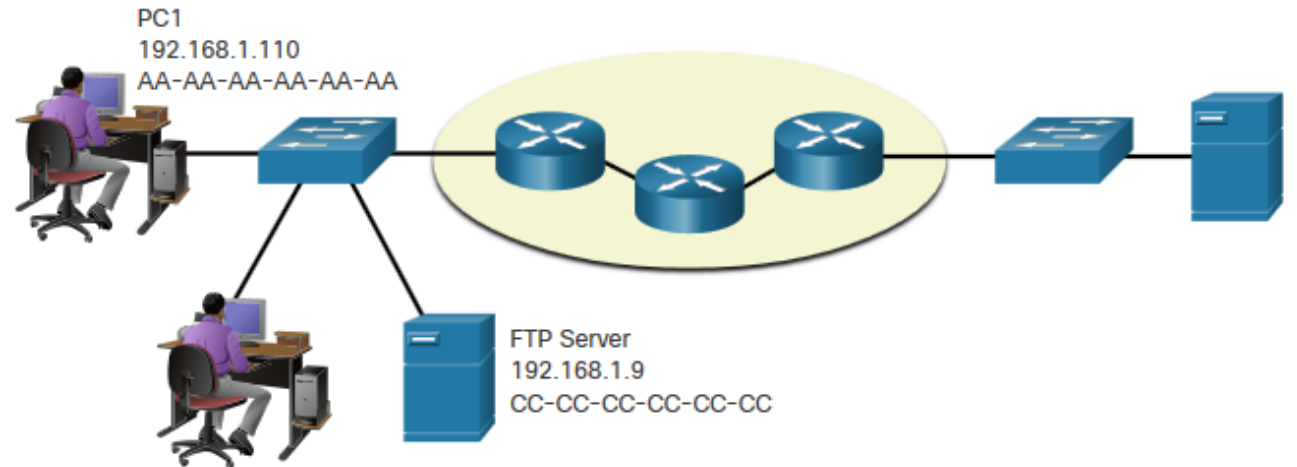
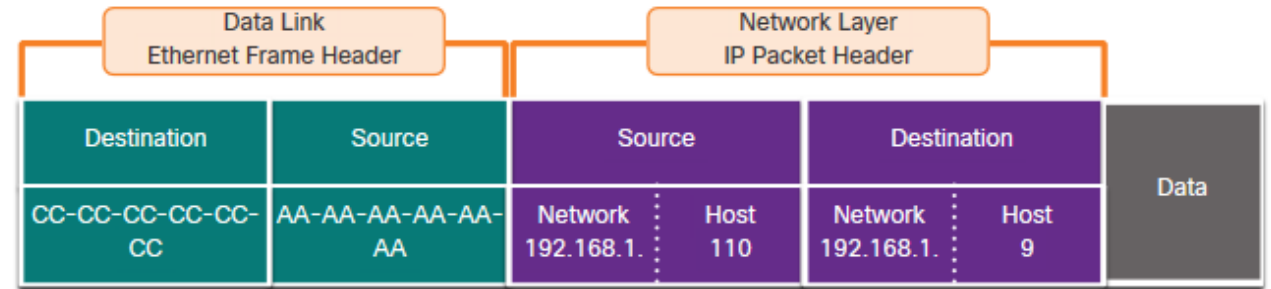
- The remaining part of the address identifies a specific device within the group.
- This portion is unique for each device on the network.



Devices on the Same Network

When devices are on the same network the source and destination will have the same number in network portion of the address.

- PC1 – 192.168.1.110
- FTP Server – 192.168.1.9

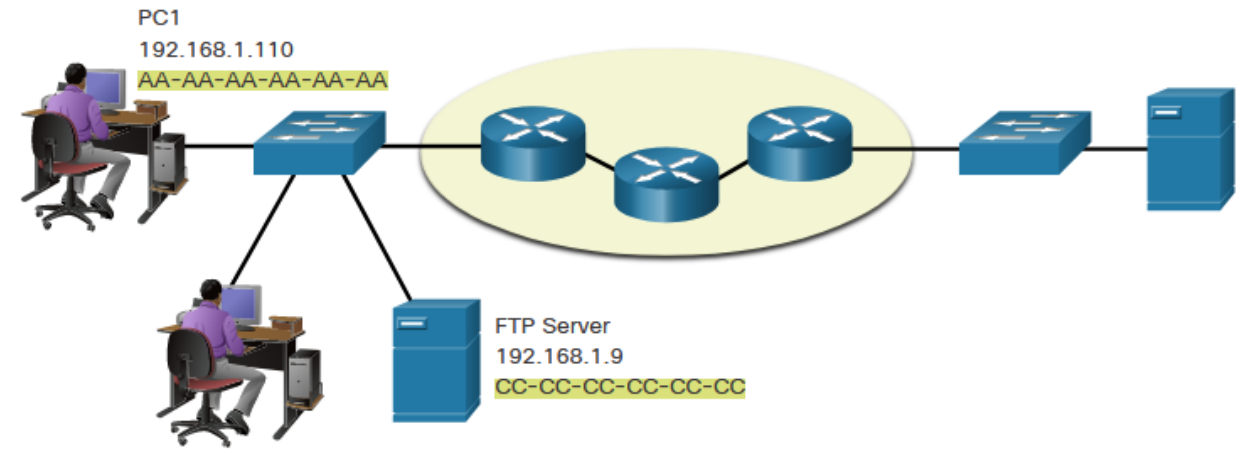
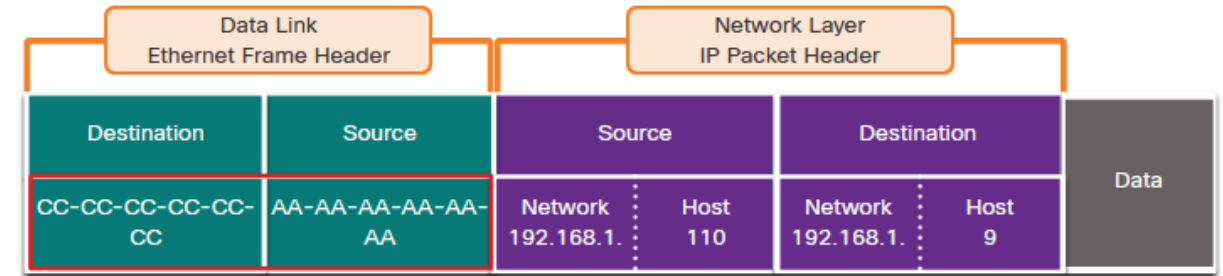


Data Link Layer Addresses: Same IP Network

When devices are on the same Ethernet network the data link frame will use the actual MAC address of the destination NIC.

MAC addresses are physically embedded into the Ethernet NIC and are local addressing.

- The Source MAC address will be that of the originator on the link.
- The Destination MAC address will always be on the same link as the source, even if the ultimate destination is remote.

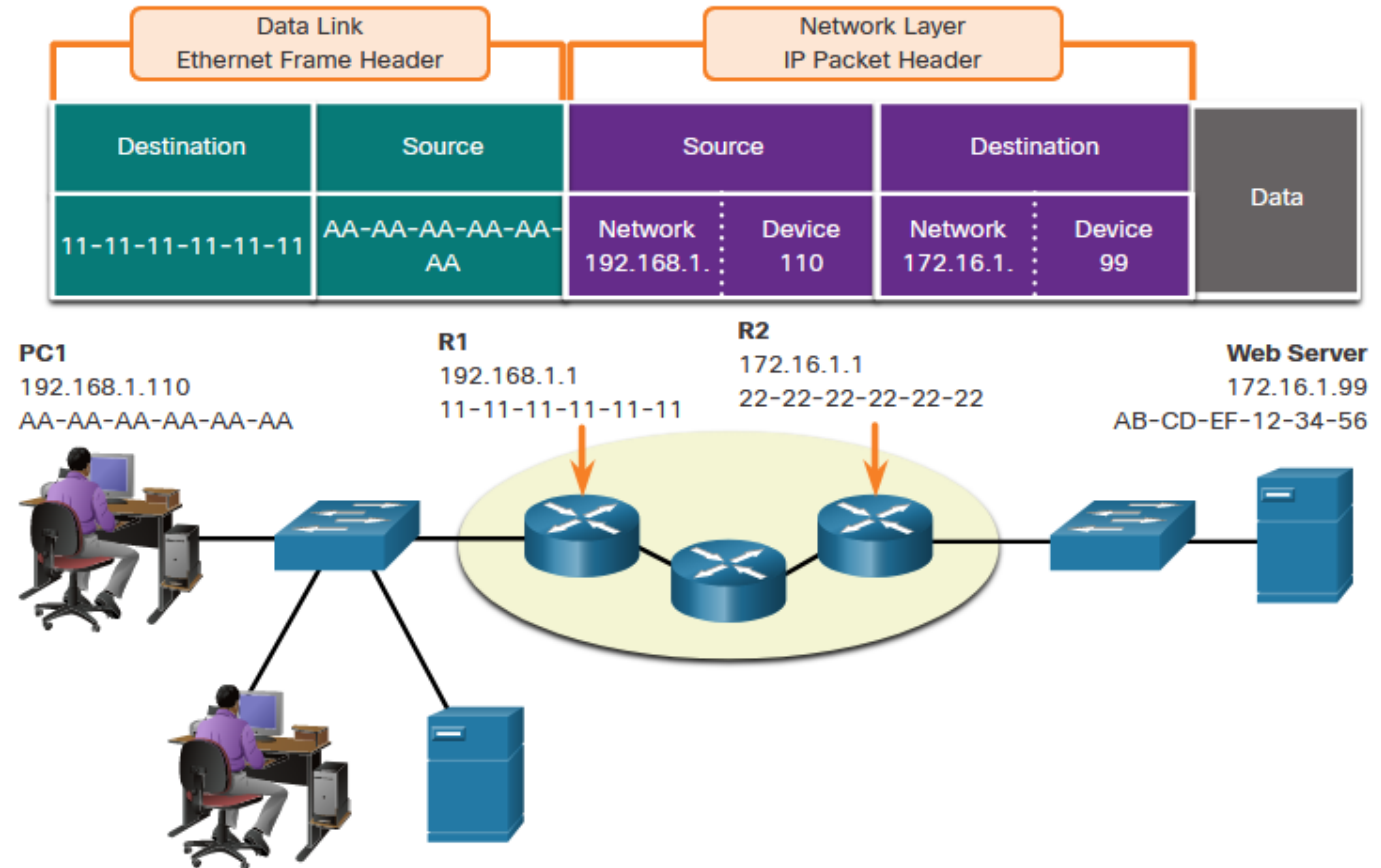


Devices on a Remote Network

- What happens when the actual (ultimate) destination is not on the same LAN and is remote?
- What happens when PC1 tries to reach the Web Server?
- Does this impact the network and data link layers?

When the source and destination have a different network portion, this means they are on different networks.

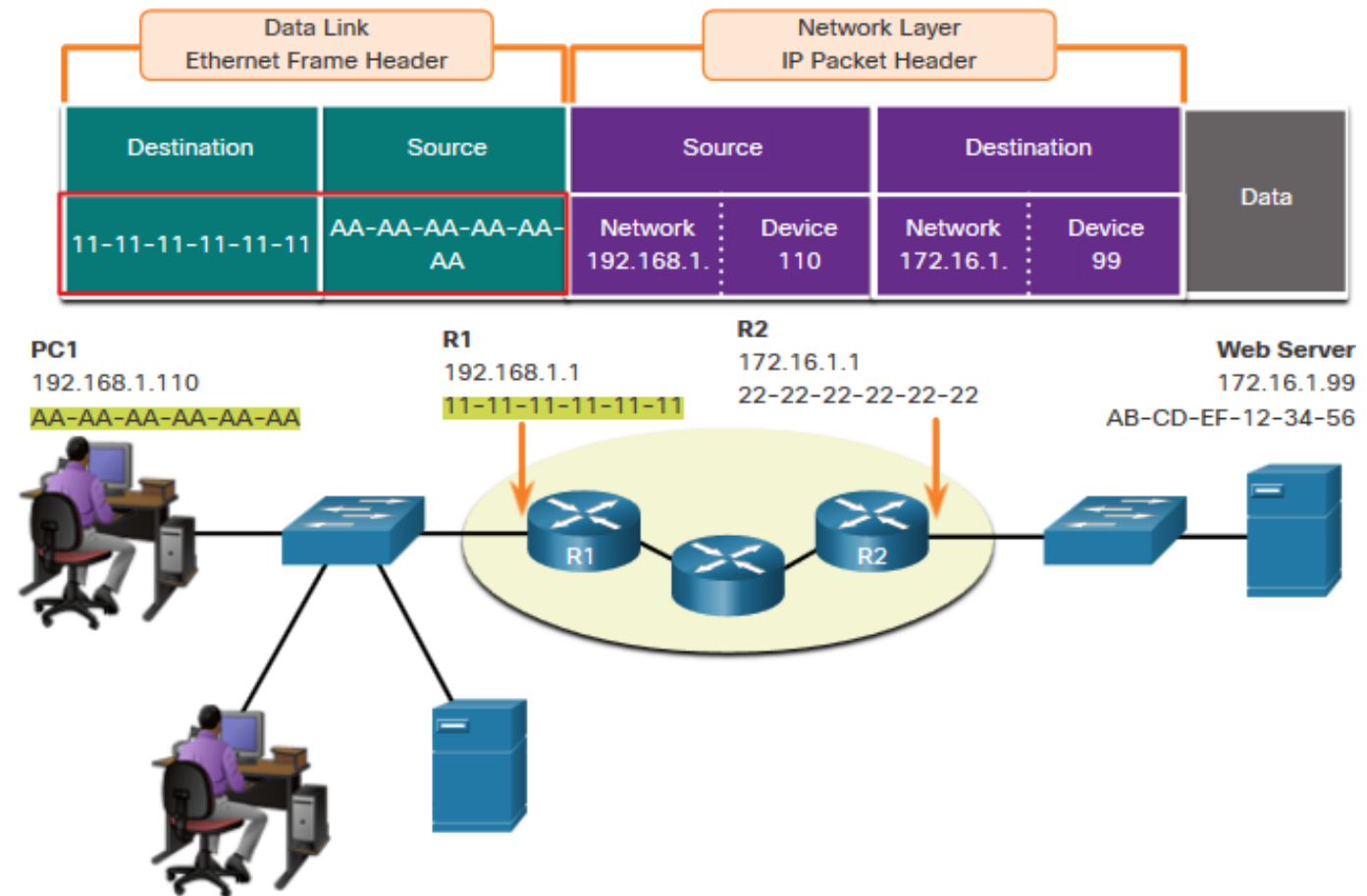
- PC1 – 192.168.1
- Web Server – 172.16.1



Role of the Data Link Layer Addresses: Different IP Networks

When the final destination is remote, Network Layer(Layer 3) will provide Data Link Layer(Layer 2) with the local default gateway IP address, also known as the router address.

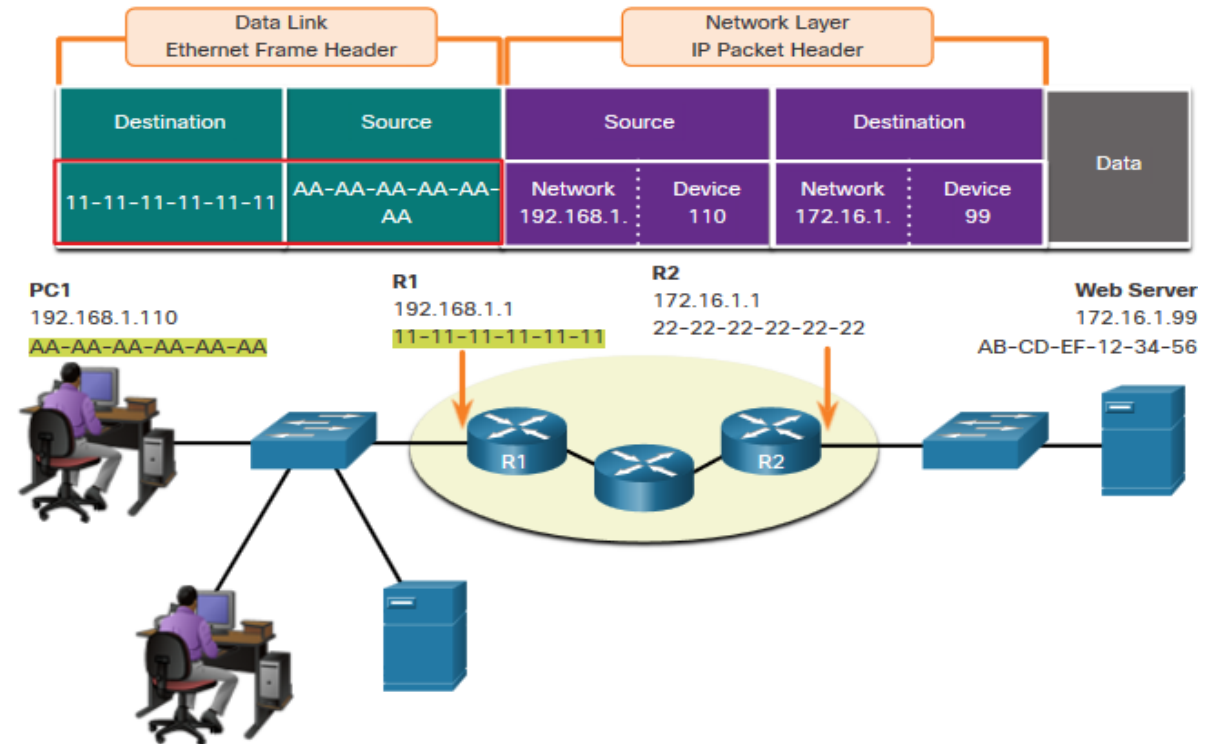
- The **default gateway (DGW)** is the router interface IP address that is part of this LAN and will be the “door” or “gateway” to all other remote locations.
- **All devices on the LAN must be told about this address or their traffic will be confined to the LAN only.**
- Once Layer 2 on PC1 forwards to the default gateway (Router), the router then can start the routing process of getting the information to actual destination.



Role of the Data Link Layer Addresses: Different IP Networks (Cont.)

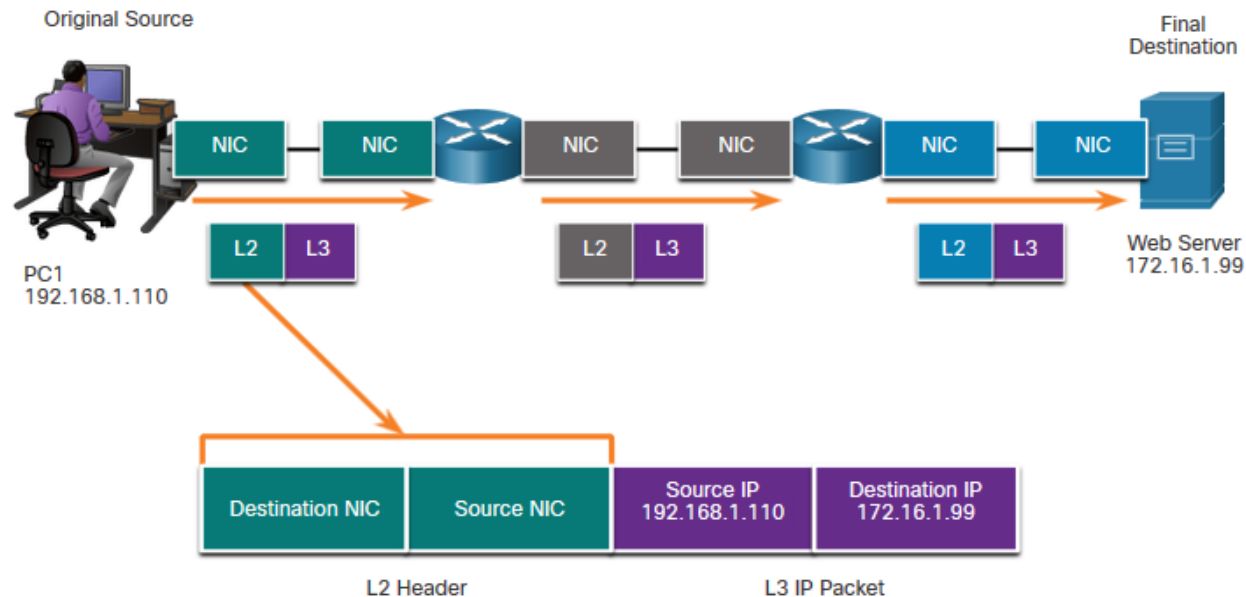
- The data link addressing is local addressing so it will have a source and destination for each link.
- The MAC addressing for the first segment is :
 - Source – AA-AA-AA-AA-AA-AA (PC1) Sends the frame.
 - Destination – 11-11-11-11-11-11 (R1-**Default Gateway MAC**) Receives the frame.

Note: While the L2 local addressing will change from link to link or hop to hop, the L3 addressing remains the same.



Data Link Addresses

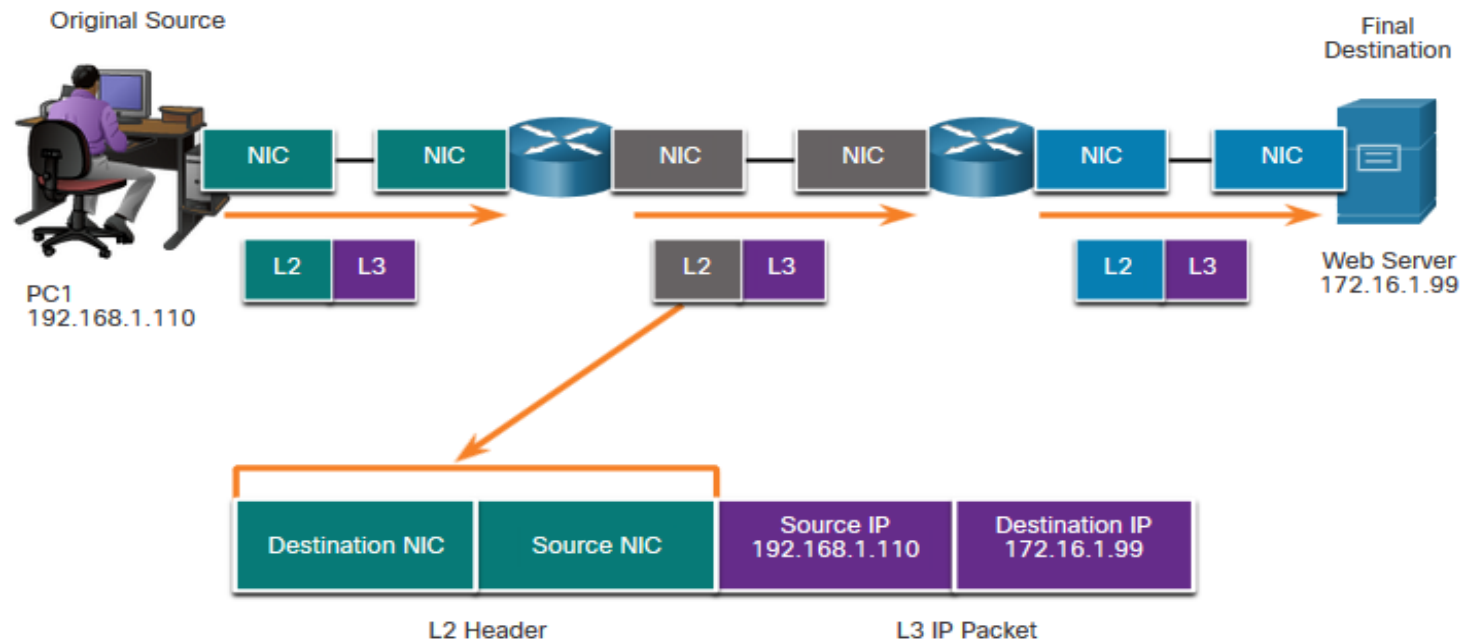
- Since data link addressing is local addressing, it will have a source and destination for each segment or hop of the journey to the destination.
- The MAC addressing for the first segment is:
 - Source – (PC1 NIC) sends frame
 - Destination – (First Router- DGW interface) receives frame



Data Link Addresses (Cont.)

The MAC addressing for the second hop is:

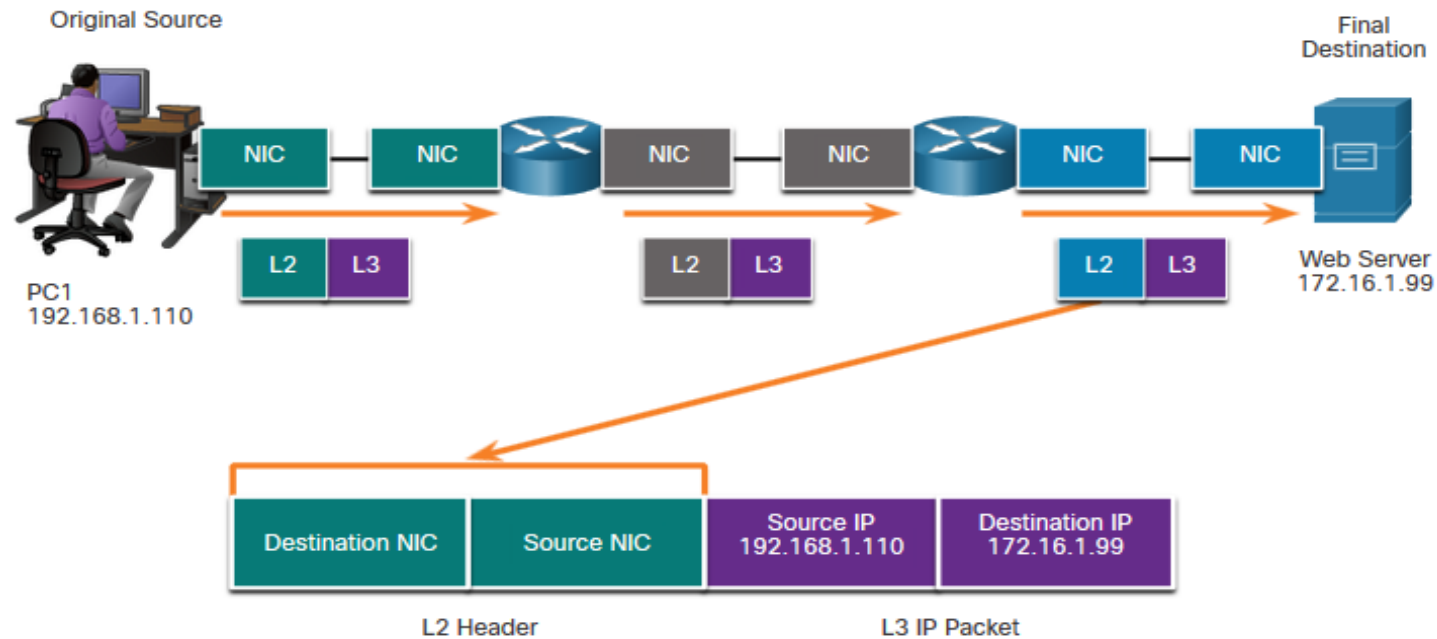
- Source – (First Router- exit interface) sends frame
- Destination – (Second Router) receives frame



Data Link Addresses (Cont.)

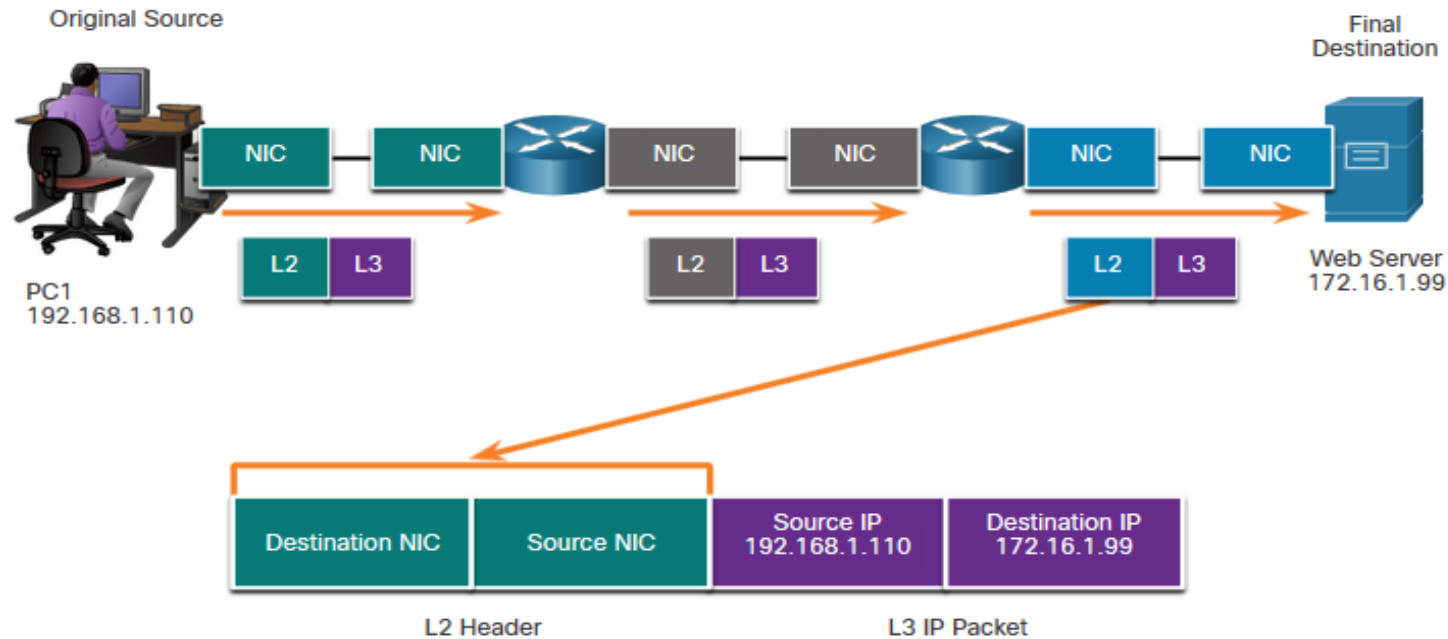
The MAC addressing for the last segment is:

- Source – (Second Router- exit interface) sends frame
- Destination – (Web Server NIC) receives frame

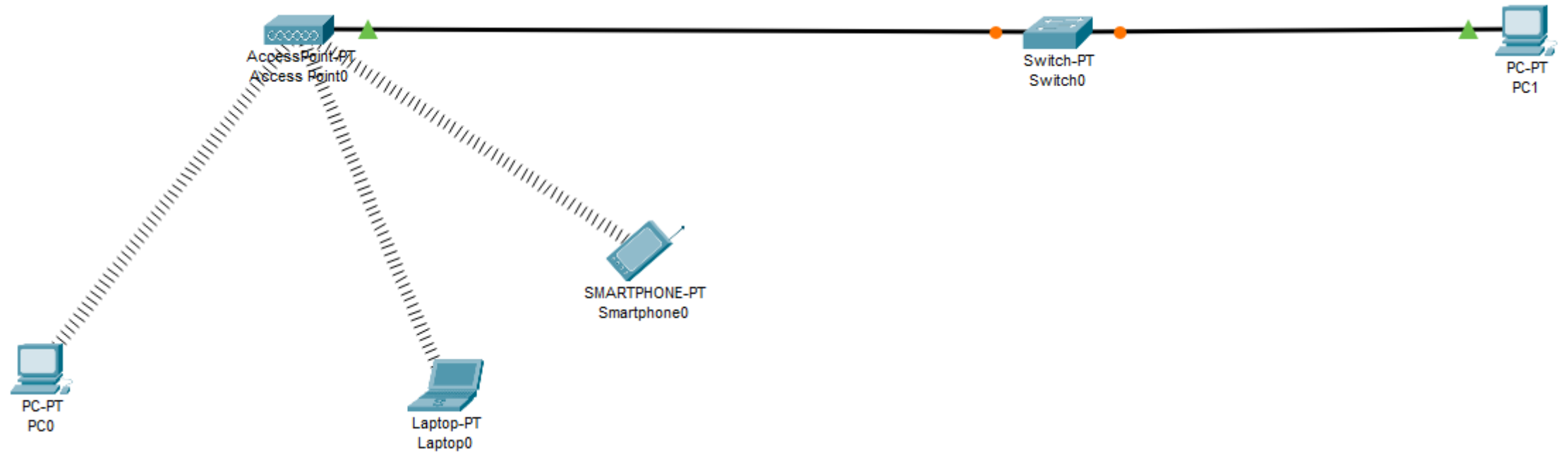


Data Link Addresses (Cont.)

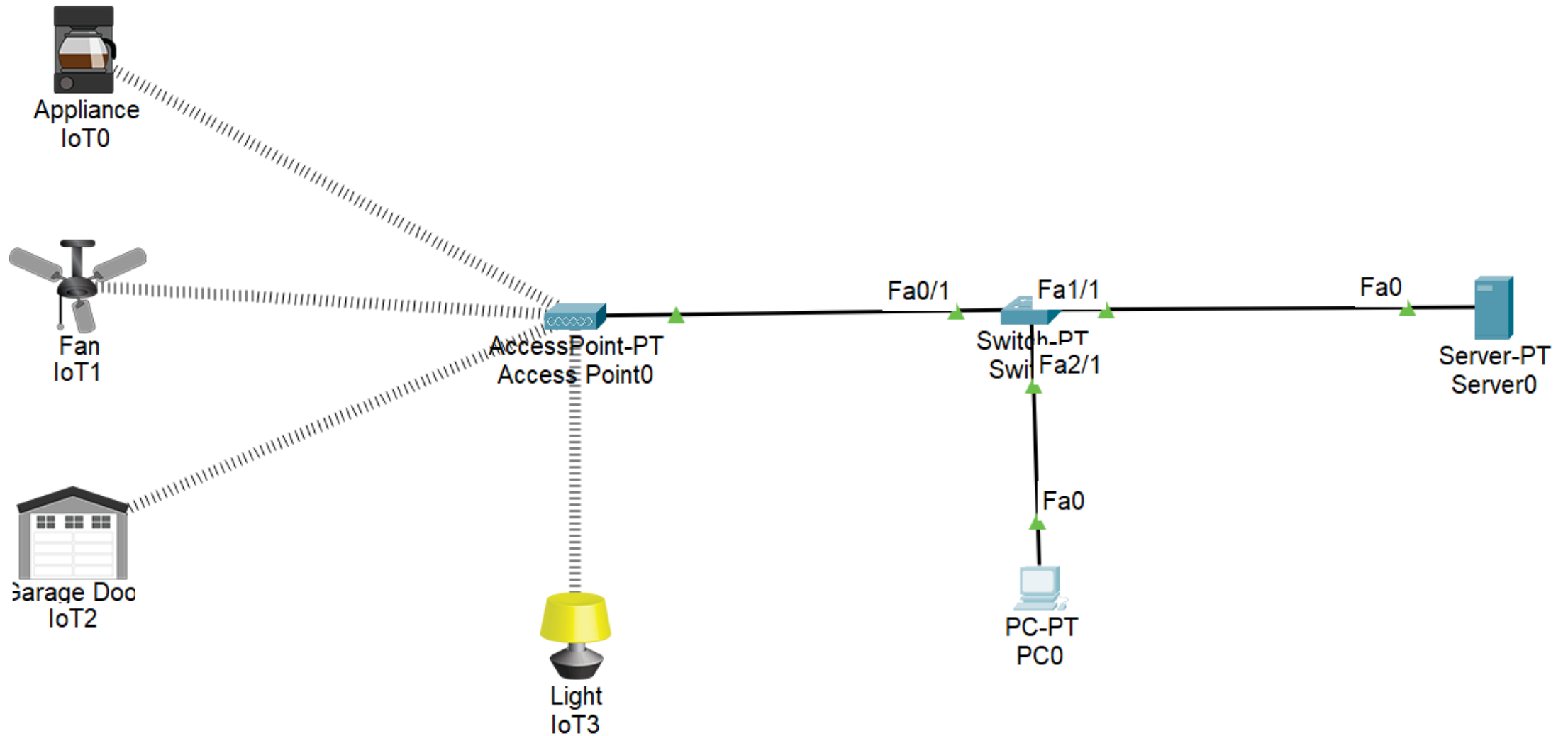
- Notice that the packet is not modified, but the frame is changed, therefore the L3 IP addressing does not change from segment to segment like the L2 MAC addressing.
- The L3 addressing remains the same since it is global and the ultimate destination is still the Web Server.



Connect IoT Devices using Switch



Connect and Manage IoT Devices



Server0

Physical Config Services **Desktop** Programming Attributes

IoT Monitor X

Registration Server Login

Wrong username or password

Username:

Password:

Don't have an IoE account? [Sign up now](#)

Server0

Physical Config Services **Desktop** Programming Attributes

IoT Monitor X

Registration Server Account Creation

Username:

Password:

Quiz

Which of the following devices operate primarily at the Data Link Layer(OSI Layer 2)

1. Router
2. Web Server
3. Switch
4. Laptop

Which of the following statements are True for communication between different networks (ie across the internet)

1. The source and destination MAC addresses will remain the same from link to link
2. The destination IP address will change from link to link.
3. The source and destination MAC address will change from link to link/hop to hop
4. The destination IP address will remain the same across all links.

Break!!!